ECE 310 Fall 2026
Lecture 18: Ideal Digital-to-Analog Conversion

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University of Illinois Urbana-Champaign

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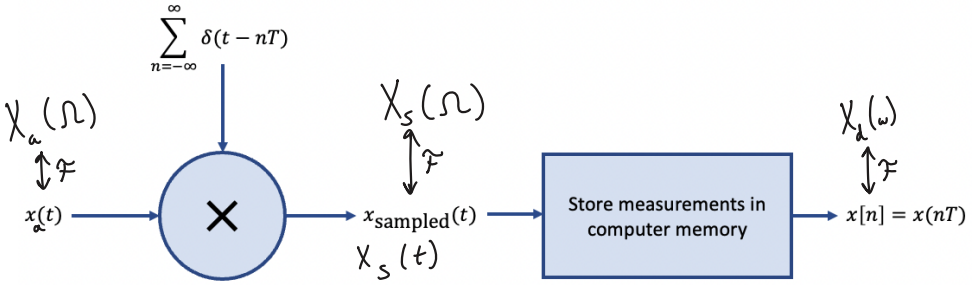
Learning Objectives

After this lecture, you should be able to:

- Explain ideal digital-to-analog conversion using sinc interpolation.
- Describe ideal digital-to-analog conversion in the frequency domain via ideal low-pass filtering.



Recap of Ideal Analog-to-Digital Conversion



$$X_s(\Omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X_a\left(\Omega - \frac{2\pi}{T}k\right)$$

$$X_d(\omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X_a\left(\frac{\omega - 2\pi k}{T}\right)$$

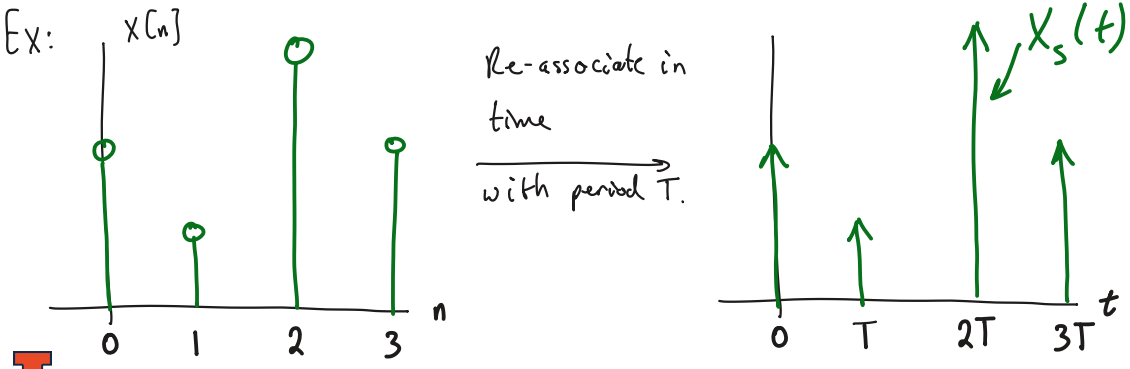
$$\omega = \Omega T$$

How can we reconstruct $x_a(t)$ from $x[n]$?

For a given sampling period T, we may re-associate samples from $x[n]$ in time:

$$x[n] = x_a(nT).$$

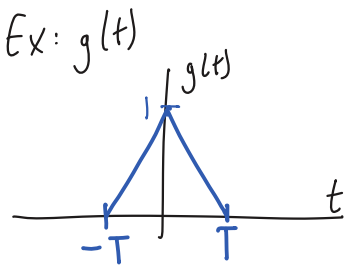
What is $x_a(t)$ when $t \neq nT$?

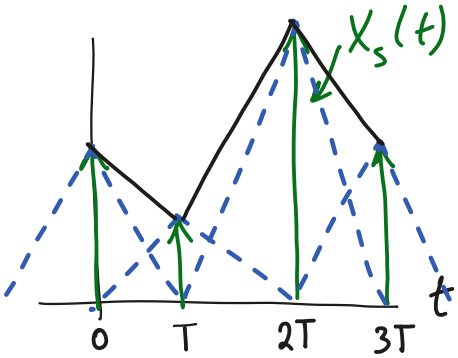


Interpolation

The process of "filling in the blanks" or "connecting the dots" is known as *interpolation*. The interpolation of digital samples to recover an analog signal can be expressed using the following reconstruction formula.

$$x_a(t) = \sum_{n=-\infty}^{\infty} x[n]g(t - nT)$$

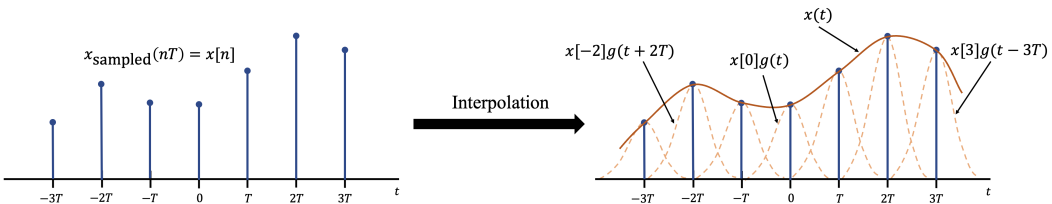




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$$x_a(t) = \sum_{n=-\infty}^{\infty} x[n]g(t - nT)$$



Ideal Digital-to-Analog Conversion

What is the correct choice for $g(t)$ to perfectly recover bandlimited analog signals?

Given $x_a(t)$ is bandlimited

$$x_a(t) = \sum_{n=-\infty}^{\infty} x[n]\, g(t-nT)$$

$$= \sum_{n=-\infty}^{\infty} x_s(nT)\, g(t-nT)$$

$$\overset{*}{=} \int_{-\infty}^{\infty} x_s(\tau)\, g(t-\tau)\, d\tau$$

$$x_s(\tau) \overset{*}{=} \begin{cases} x[n], & \tau = nT \\ 0, & \text{else} \end{cases}$$

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Ideal Digital-to-Analog Conversion

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$$\overset{*}{=} \int_{-\infty}^{\infty} x_s(\tau)\, g(t-\tau)$$

$$= x_s(t) * g(t)$$

$\updownarrow \mathcal{F}$

$$X_a(\Omega) = X_s(\Omega)\, G_a(\Omega)$$

$$x_s(\tau) \overset{*}{=} \begin{cases} x[n], & \tau = nT \\ 0, & \text{else} \end{cases}$$

Recall: $\rightarrow x_s(t) = \begin{cases} x[n] = x_a(nT), & t = nT \\ 0, & t \neq nT \end{cases}$

$\rightarrow X_s(\Omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X_a\left(\Omega - \frac{2\pi}{T}k\right)$

Ideal Digital-to-Analog Conversion

What is the correct choice for $g(t)$ to perfectly recover bandlimited analog signals?

$$X_a(\Omega) = X_s(\Omega)\, G_a(\Omega)$$

What we want:

$$X_s(\Omega) \rightarrow \boxed{G_a(\Omega)} \rightarrow X_a(\Omega)$$

$$\rightarrow x_s(t) = \begin{cases} x[n] = x_a(nT), & t = nT \\ 0, & t \neq nT \end{cases}$$

$$\rightarrow x_a(t) = \sum_{n=-\infty}^{\infty} x[n]\, g(t - nT)$$

$$\rightarrow X_s(\Omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X_a\left(\Omega - \frac{2\pi}{T} k\right)$$

$$X_a(\Omega) = \underbrace{\frac{1}{T} \sum_{k=-\infty}^{\infty} X_a\left(\Omega - \frac{2\pi}{T} k\right)}_{X_s(\Omega)} G_a(\Omega)$$



Ideal Digital-to-Analog Conversion

What is the correct choice for $g(t)$ to perfectly recover bandlimited analog signals?

$$X_a(\Omega) = X_s(\Omega)\, G_a(\Omega)$$

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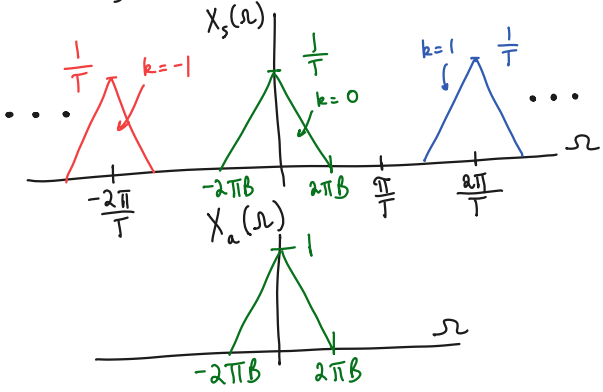
$$\rightarrow x_a(t) = \sum_{n=-\infty}^{\infty} x[n]\, g(t - nT)$$

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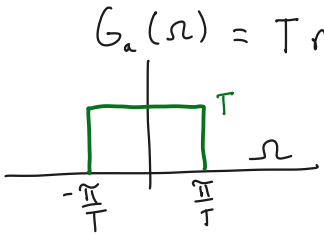
What we want:

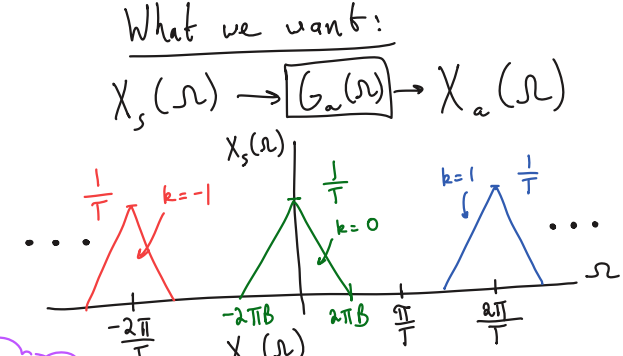
$$X_s(\Omega) \rightarrow \boxed{G_a(\Omega)} \rightarrow X_a(\Omega)$$



Ideal Digital-to-Analog Conversion

What is the correct choice for $g(t)$ to perfectly recover bandlimited analog signals?

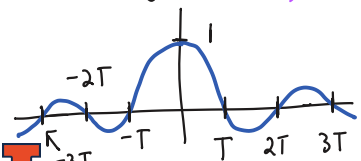




$\Longleftrightarrow$

$\updownarrow \mathcal{F}$

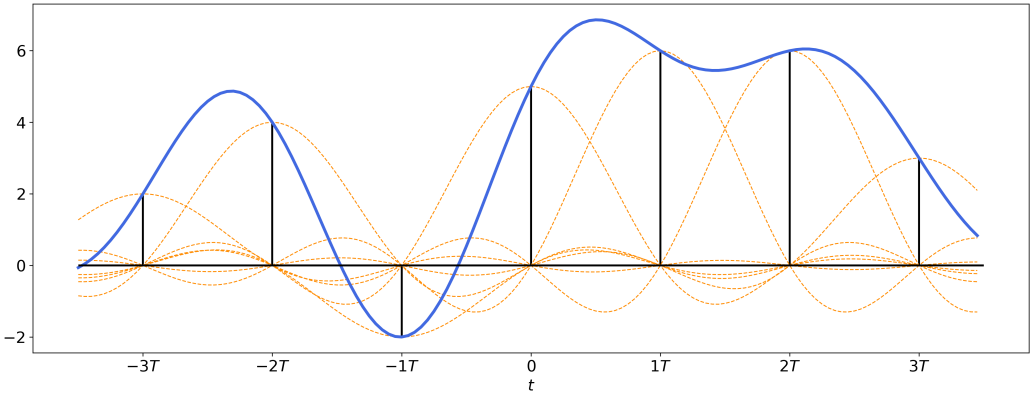
$g(t) = \mathrm{sinc}\left(\frac{\pi}{T}t\right)$



$$\mathrm{sinc}(x) = \begin{cases} 1, & x = 0 \\ \frac{\sin(x)}{x}, & x \neq 0 \end{cases}$$

Ideal Sinc Interpolation: Time-domain

$$x(t) = \sum_{n=-\infty}^{\infty} x[n]\text{sinc}\left(\frac{\pi}{T}(t - nT)\right)$$

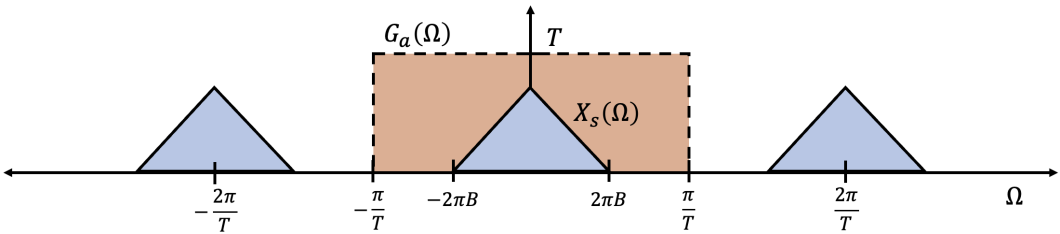


Ideal Sinc Interpolation: Frequency-domain

$$X_a(\Omega) = X_s(\Omega) G_a(\Omega)$$

$$X_s(\Omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X_a\left(\Omega - \frac{2\pi}{T} k\right)$$

$$G_a(\Omega) = T \operatorname{rect}\left(\frac{\Omega}{\frac{2\pi}{T}}\right)$$



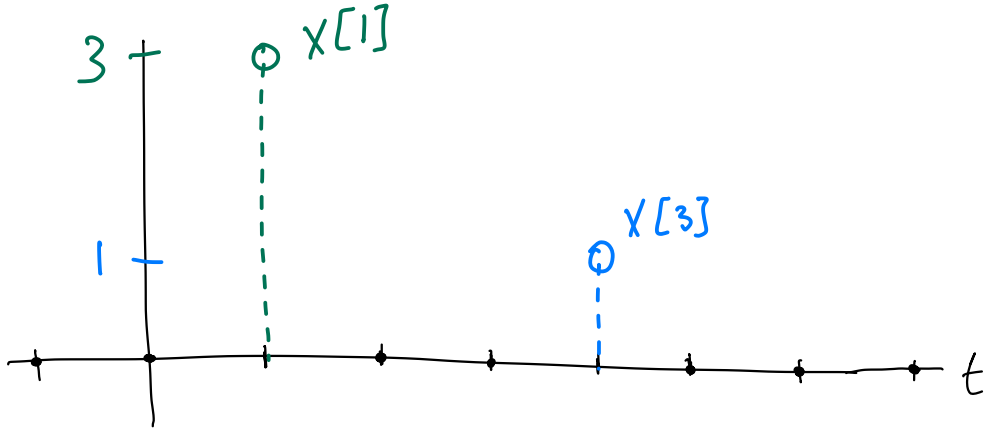
Example: Sketching Ideal D/A Conversion

Let $x[n] = 3\delta[n-1] + \delta[n-4]$. Sketch $x_a(t)$ if we recover $x[n]$ by ideal sinc interpolation with sampling period $T = 2$ seconds.



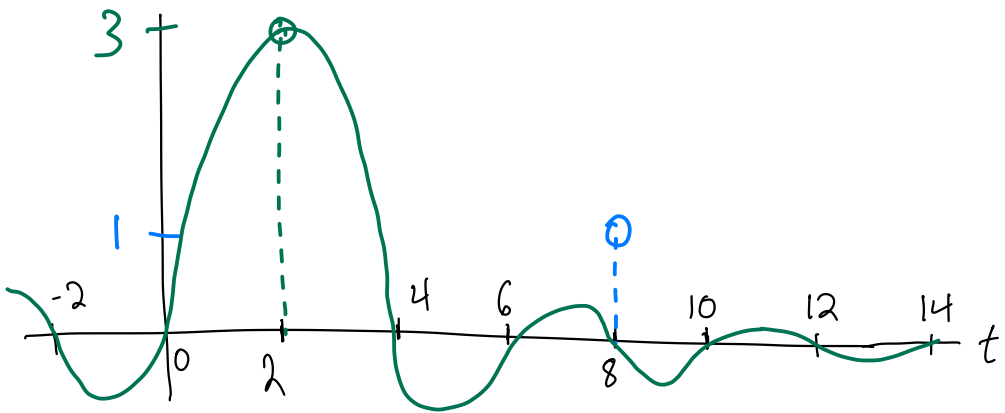
Exercise 1: Sketching Ideal D/A Conversion

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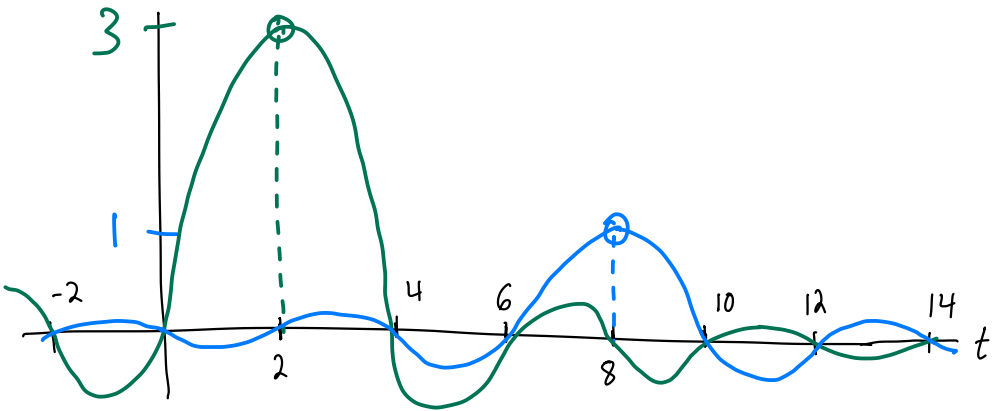
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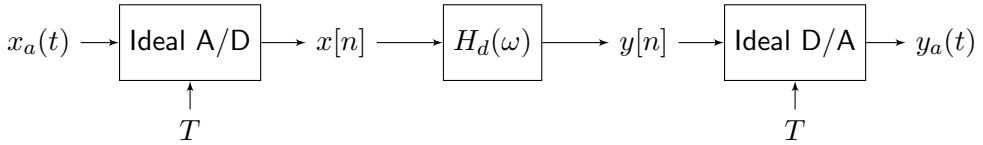
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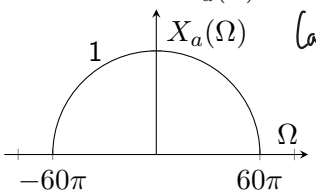


Overall Sampling System Example

Consider the below sampling system:



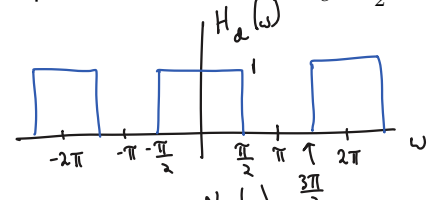
We have that $H_d(\omega)$ is a low-pass filter with cutoff $\omega_c = \frac{\pi}{2}$ and $T = \frac{1}{80}$s. $\omega = \Omega T$

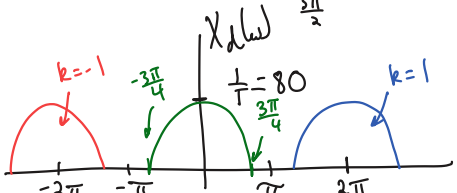


(a) Sketch $X_d(\omega)$.

(b) Sketch $Y_d(\omega)$.

(c) Sketch $Y_a(\Omega)$.





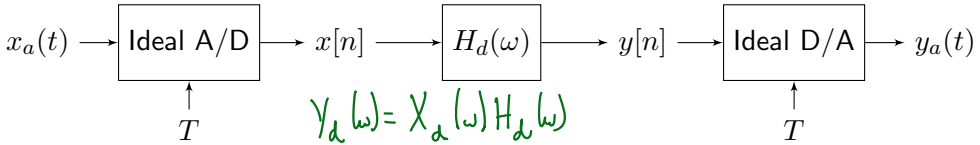
$\omega_{max} = \Omega_{max} T$

$= 60\pi \cdot \frac{1}{80}$

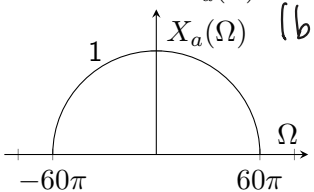
$= \frac{3\pi}{4}$

Overall Sampling System Example

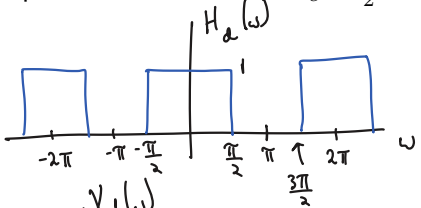
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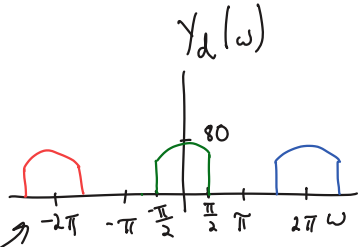


We have that $H_d(\omega)$ is a low-pass filter with cutoff $\omega_c = \frac{\pi}{2}$ and $T = \frac{1}{80}$s. $\omega = \Omega T$



(b)



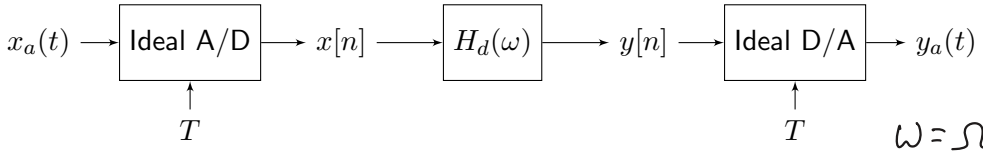


(a) Sketch $X_d(\omega)$.
(b) Sketch $Y_d(\omega)$.
(c) Sketch $Y_a(\Omega)$.

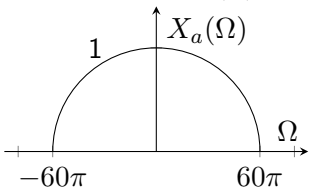


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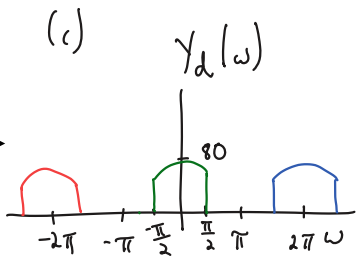
We have that $H_d(\omega)$ is a low-pass filter with cutoff $\omega_c = \frac{\pi}{2}$ and $T = \frac{1}{80}$s.



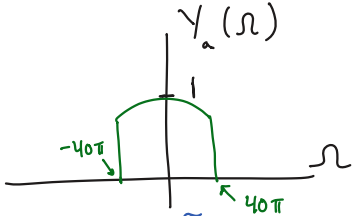
(a) Sketch $X_d(\omega)$.

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(c) Sketch $Y_a(\Omega)$.



D/A



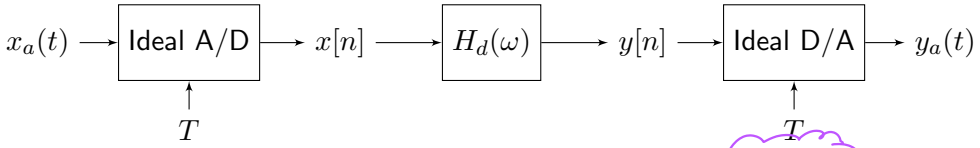
$\omega = \Omega T$

$\Omega = \frac{\omega}{T}$

$\Omega_{max} = \frac{\omega_{max}}{T} = \frac{\frac{\pi}{2}}{\frac{1}{80}} = 40\pi$

Overall Sampling System Example

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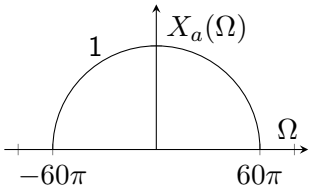


We have that $H_d(\omega)$ is a low-pass filter with cutoff $\omega_c = \frac{\pi}{2}$ and $T = \frac{1}{50}$s.

$\omega = \Omega T$

$\omega_{max} = 60\pi \cdot \frac{1}{50} = \frac{6\pi}{5}$

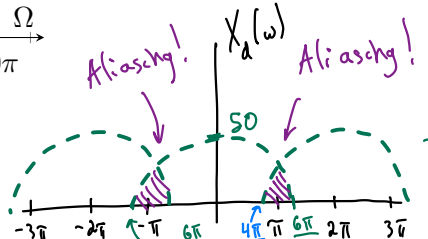
New T!

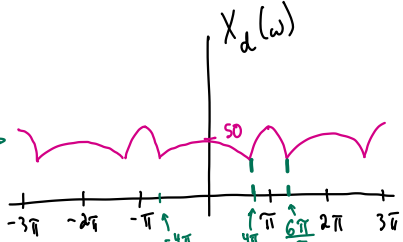


(a) Sketch $X_d(\omega)$.

(b) Sketch $Y_d(\omega)$.

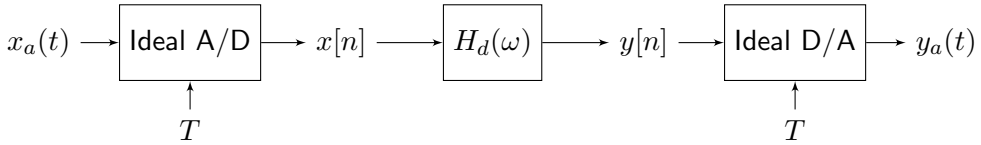
(c) Sketch $Y_a(\Omega)$.



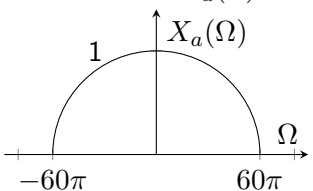


Overall Sampling System Example

Consider the below sampling system:



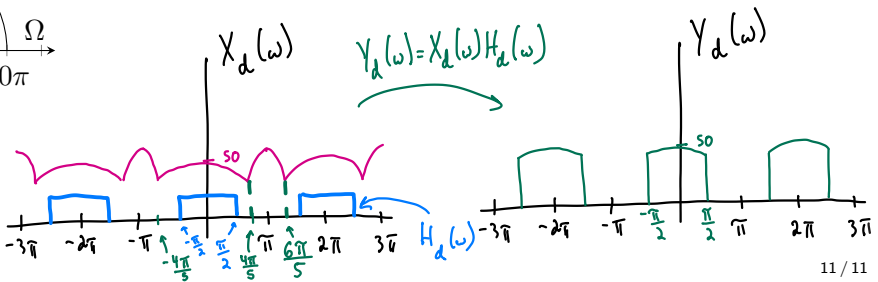
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(a) Sketch $X_d(\omega)$.

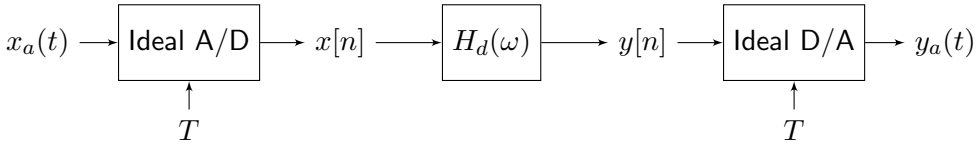
(b) Sketch $Y_d(\omega)$.

(c) Sketch $Y_a(\Omega)$.

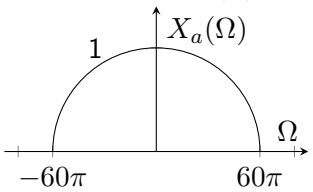


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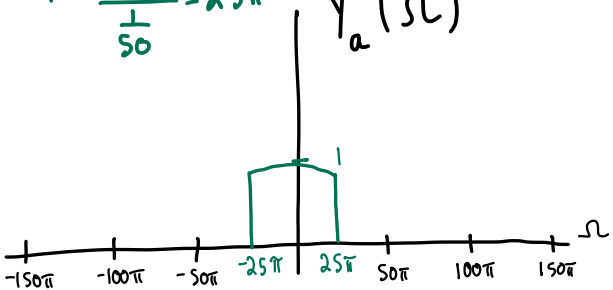


We have that $H_d(\omega)$ is a low-pass filter with cutoff $\omega_c = \frac{\pi}{2}$ and $T = \frac{1}{50}$s.



$\Omega = \frac{\omega}{T} = \frac{\left(\frac{\pi}{2}\right)}{\frac{1}{50}} = 25\pi$

$Y_a(\Omega)$



(a) Sketch $X_d(\omega)$.

(b) Sketch $Y_d(\omega)$.

(c) Sketch $Y_a(\Omega)$.